

QuickCart REST API Black-Box Test Report

Date: 23 March 2026

Testing Method: Black-box API testing using pytest and requests

Scope: Validation of HTTP status codes, JSON structure, and behavior against API documentation

Executive Summary

Total verified bug test cases: 17

Total distinct logical bugs: 11

The API was tested with valid requests, invalid inputs, missing/incorrect field values, wrong data types, and boundary conditions. Major issues were found in cart math, checkout rules, stock restoration, and input validation.

Environment and Approach

- Base URL: `http://localhost:8080/api/v1`
- Automated test suite location: `blackbox/tests/`
- Coverage areas: auth, profile, products, cart, checkout, orders, wallet, addresses, reviews, coupons, loyalty, support

Bug Reports

BUG-01: Cart accepts zero and negative quantities (High)

Endpoint: `POST /cart/add`

Expected: Quantity must be ≥ 1 , reject invalid values with 400.

Actual: API accepts 0 and negative values and returns 200 OK.

BUG-02: Cart item subtotal calculation is incorrect (High)

Endpoint: GET /cart

Expected: Item subtotal = quantity x unit_price.

Actual: Returned subtotal is mathematically inconsistent.

BUG-03: Cart total calculation is incorrect (High)

Endpoint: GET /cart

Expected: Cart total = sum of item subtotals.

Actual: Returned total does not match item subtotals.

BUG-04: Checkout allows COD for orders over Rs. 5000 (High)

Endpoint: POST /checkout

Expected: COD must be rejected for totals above Rs. 5000.

Actual: Order placed successfully with COD and 200 OK.

BUG-05: WALLET checkout succeeds with insufficient balance (High)

Endpoint: POST /checkout

Expected: Checkout fails when wallet funds are insufficient.

Actual: Order succeeds with 200 OK despite low/empty wallet.

BUG-06: Order cancellation does not restore product stock (High)

Endpoint: POST /orders/{order_id}/cancel

Expected: Cancelled quantities should be returned to stock.

Actual: Order cancels, but stock remains unchanged.

BUG-07: Addresses API accepts non-numeric pincodes (Medium)

Endpoint: POST /addresses

Expected: Pincode must be exactly 6 digits.

Actual: Alphabetic values are accepted.

BUG-08: Profile accepts invalid phone format (Medium)

Endpoint: PUT /profile

Expected: Phone must be exactly 10 digits.

Actual: Alphanumeric phone values accepted with 200 OK.

BUG-09: Reviews accept ratings outside 1-5 (Medium)

Endpoint: POST /products/{product_id}/reviews

Expected: Rating must be integer between 1 and 5.

Actual: Ratings such as 0, -1, 6, and 100 are accepted.

BUG-10: Wallet deduction precision is inaccurate (High)

Endpoint: POST /wallet/pay

Expected: Balance should be reduced by exact payment amount.

Actual: Floating-point/rounding issues produce incorrect final balance.

BUG-11: Non-existent user ID returns 404 instead of documented 400 (Low)

Endpoint: User-scoped endpoints (example: GET /profile)

Expected: Invalid/non-existent X-User-ID should return 400 Bad Request.

Actual: API returns 404 Not Found.

Conclusion

The black-box suite identified critical business-rule violations and validation gaps. Highest priority fixes are

cart math correctness, checkout policy enforcement, wallet-balance checks, and stock restoration on cancellation.